Anna Sustayta

CSCI/MATH 485

Dr. Bo Shen

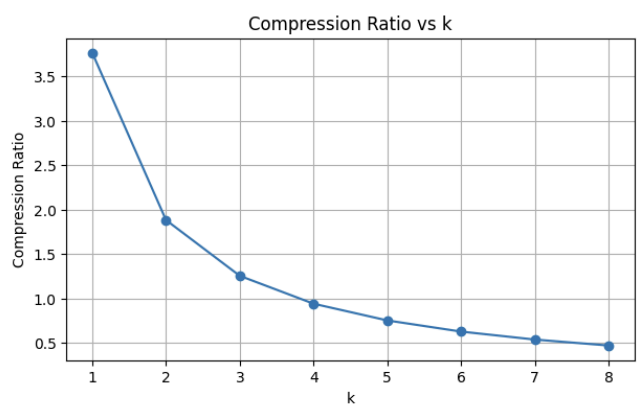
Assignment 3

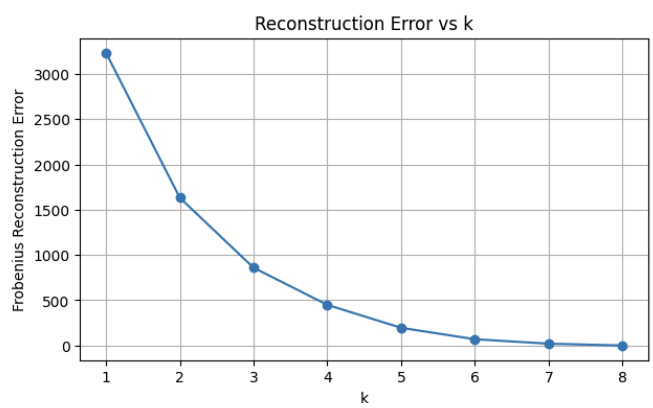
1. Introduction

Singular Value Decomposition provides an efficient and effective method for approximating matrices with lower rank representations. This method allows images to be formed into matrices to be approximate using a few parameters while preserving important visual features. Image compression serves a powerful application of linear algebra and matrix factorization techniques. This assignment will allow for the expiration of image compression using block-wise SVD. The image is converted into a gray-scale, partitioned into small blocks, and SVD is applied to each block individually. Retaining only the top k singular value, we can reconstruct the image while reducing the amount of data stored. The effect of varying k from 1 to 8 is analyzed in terms of compression ratio and reconstruction quality.

2. Methods

The selected image `Cats.jpeg` was converted into a gray-scale and resized so that its dimensions were divisible by 8. The image matrix was then partitioned into non-overlapping $8x8$ blocks. For each block B, Singular Value Decomposition was applied for $B = U\Sigma V^T$. To compress these blocks, the top-k singular values were stored for $B_K = U_k \Sigma_k V_k^{\ T}$. This allowed a rank-$k$ approximation of the reconstructed blocks, that were recombined to form the final compressed image. This process continued for values $k \in \{1, 2, 3,.., 8\}$.



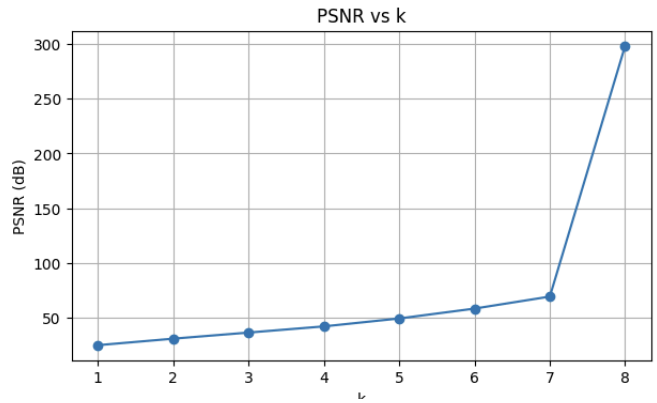


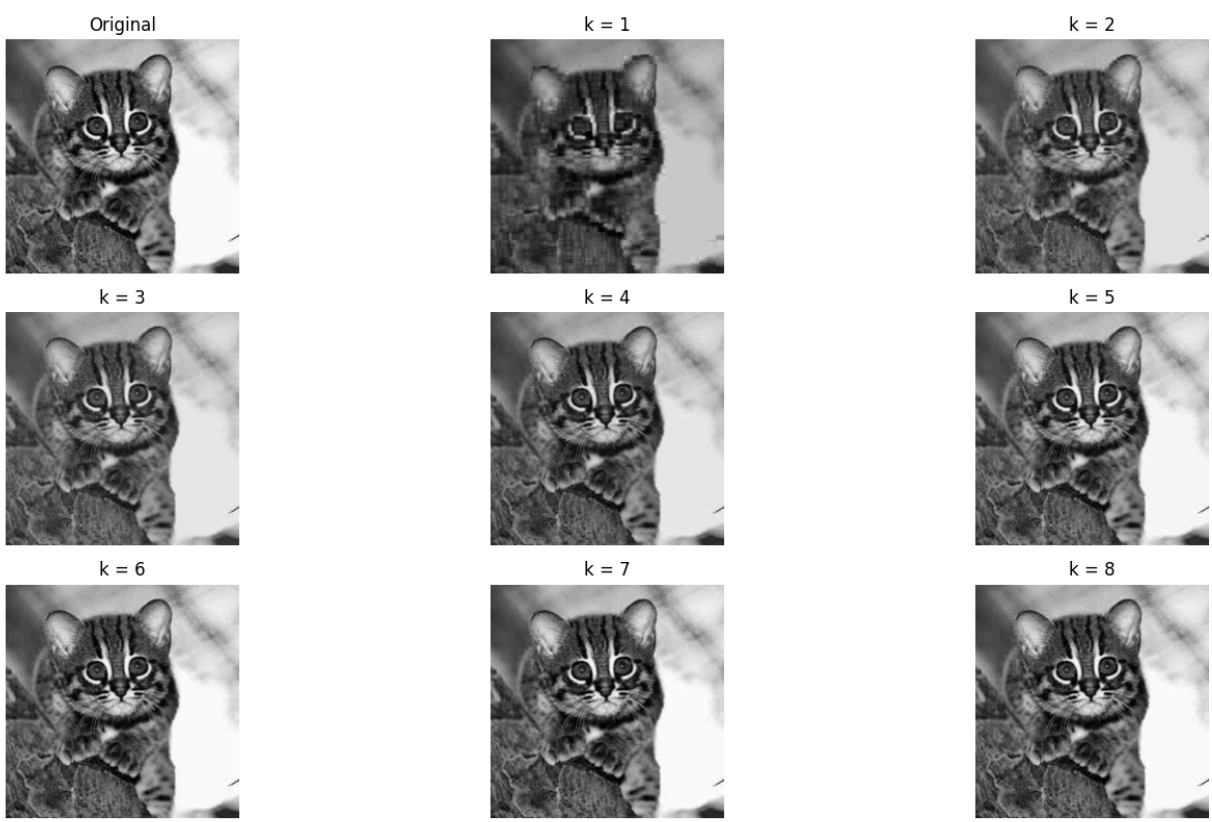
2i. Compression Ratio

The compression ratio measures how much the data size is reduced. For each $8x8$ block the original data stored 64 values. The SVD compression ratio calls for $k(U + \Sigma + V^T) = k(8 + 8 + 1)$, to get our final ratio $64/17k$. Lower values of k provide greater compression, but can result in lower reconstruction quality.

$2ii$. Reconstruction Quality

The reconstruction quality was evaluated using the Frobenius norm error $||A - A_k||_F$ in which A was used for the original image, and A_K is the reconstructed image. Peak Signal-to-Noise Ratio (PSNR) was calculated to provide a visual quality metric. The PSNR had reassured that with an increase in k , the reconstruction error decreases because more singular values are retained.



3. Results

The reconstructed images provide higher image quality as k increases, and for lower values of k, more detail is lost and the image will appear more blurry. However, with a maximum rank of 8 in each block, $k = \{4, 5\}$ show a coherent reconstruction. The compression ratio also decreases as k increases due to more information being stored, showing the trade-off between compression efficiency and image quality. Overall, SVD can be shown to be an efficient and useful method by reducing storage requirements while preserving image quality.